Introduction to Neuro-Symbolic AI

Neuro-Symbolic Artificial Intelligence

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Who am I?

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Who are you?

Background?

Experience in Machine Learning? Previous courses on logic?

Programming experience? Python?

What is this part of the course about?

Integration of **deep learning** and **symbolic** computing. How to model machine learning problems in the logic framework

Topics:

1. Kautz's Taxonomy
2. Crush course on Logic
3. Logic Tensor Networks
4. Knowledge Graph Embeddings
5. Neural SAT
6. Logic Neural Networks
7. Neural ASP

Teaching modalities:

- Frontal lectures
- Hands-on sessions (typically on the 1 hour slot on Thursdays) - BYOL

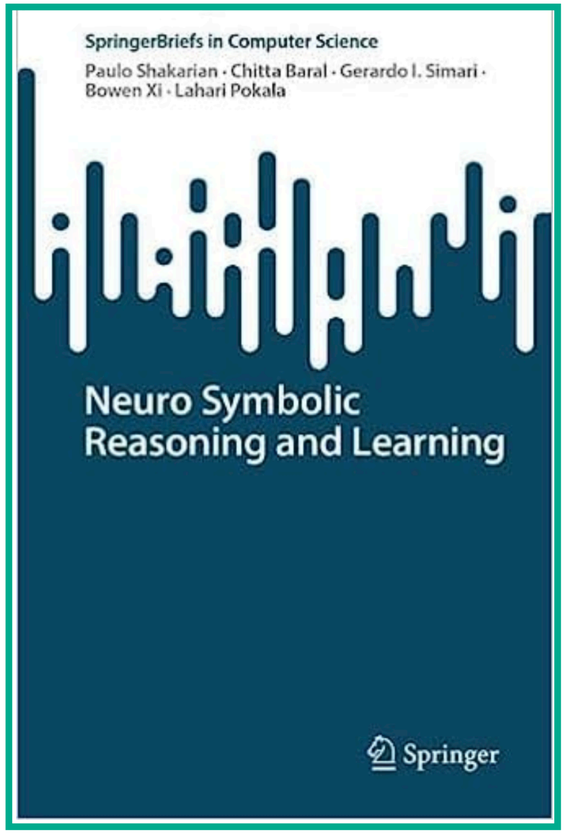
Material

Textbook and online material

- Book, slides, videos:

  https://neurosymbolic.asu.edu/advances-in-neuro-symbolic-reasoning-and-learning/

- Course material (slides, registrations, etc.) will be uploaded on Teams



Neuro Symbolic Reasoning and Learning (in press, 2023) by Paulo Shakarian, Gerardo I. Simari, Chitta Baral, Bowen Xi, and Lahari Pokala

Exam

Goals:

Being capable of *combining* several methods of *symbolic reasoning* with different *learning* paradigms to create models with improved efficiency and interpretability.

Use state of the art tools to model and solve neuro-symbolic learning problems.

Exam: Oral interview (group project must be discussed and approved before the exam)

- **Option 1:** Presentation of a **relevant paper** about neurosymbolic computing not seen in detail during the lectures
- **Option 2: Project** where the topics seen during the course are applied to a **novel problem**.
- **Option 3:** More challenging project merging concepts of XAI and NeSy part.

Introduction to Neuro-Symbolic AI

Introduction to Neuro-Symbolic AI

PERCEPTION

Latin `perceptio` ‘gathering, receiving’

Transforming raw sensory data into meaningful representations

“What am I observing?”

COGNITION

Latin `cognitio` ‘examination, learning, knowledge’

Reasoning over representations of knowledge to draw conclusions and make decisions.

“What does it mean, and what should I do?”

Example: A robot perceives an obstacle in front of it (perception) and decides to move around it (cognition).

Humans interact with the environment using a combination of **perception** and **cognition** for supporting abstraction, reasoning by analogy, and long-term planning.

Humans can control and explain their cognitive functions! For example, humans can follow and explain the guidelines and safety constraints driving their decision-making in safety-critical applications such as healthcare, criminal justice, and autonomous driving.

Neuro-Symbolic AI

Explaininig cognitive functions

Retention of symbolic mappings from perception outputs to knowledge about their environment.

Data-driven neural network-based AI algorithms effectively model **machine perception**

Symbolic knowledge-based AI better suited for modeling **machine cognition**.

Symbolic knowledge structures support explicit representations of mappings from perception outputs to the knowledge, enabling traceability and auditing of the AI system's decisions.

Neuro-Symbolic AI

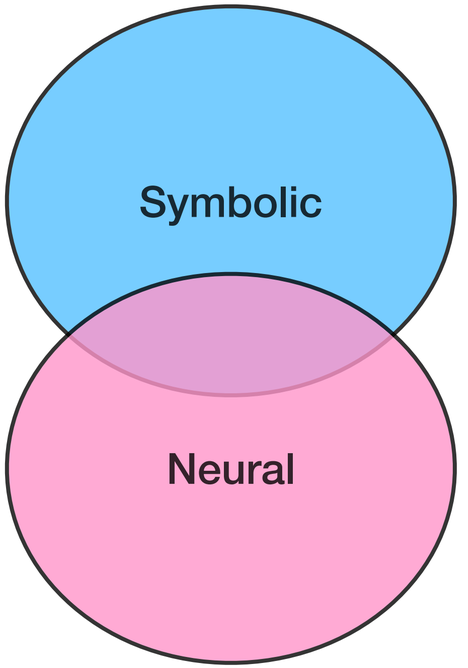
Combination of deep learning and symbolic reasoning.

The argument for this **hybrid approach** is that neural and symbolic systems can complement each other and mitigate their respective weaknesses.

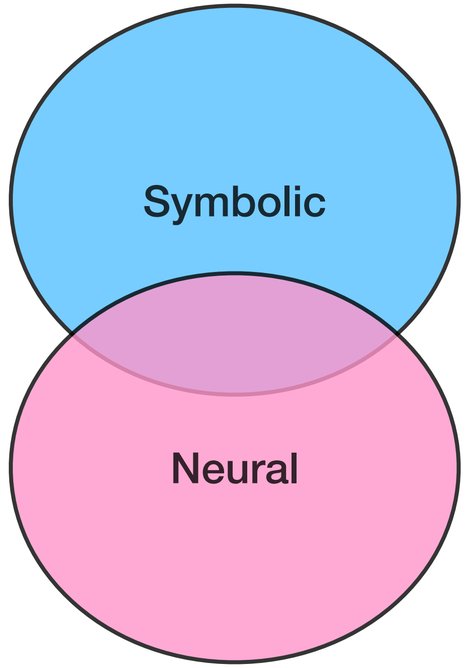
> For many applications, it is highly desirable to combine accountable and interpretable logic-based modules with effective deep learning ones.

While **interpretability** is defined in various ways across the literature, it is generally viewed as the ability to be understood by a human.

> An end-user might easily comprehend the reasoning processes that led to model predictions.



Neuro-Symbolic AI

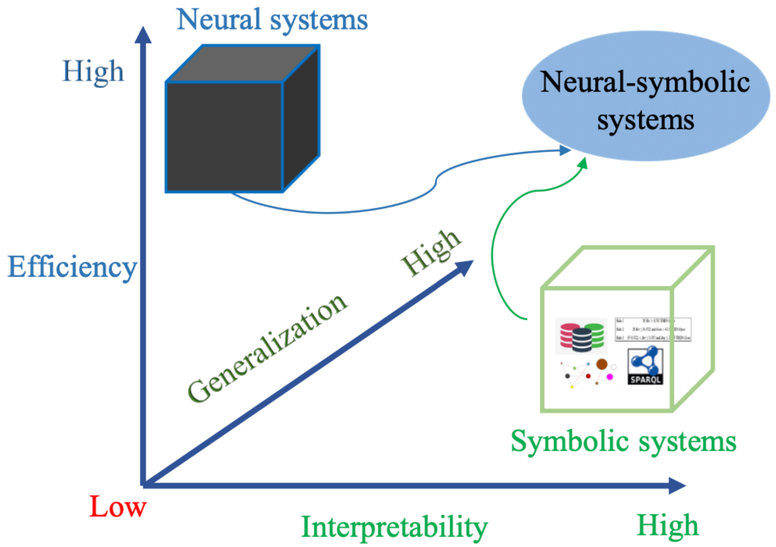


Neuro-symbolic AI refers to AI systems that seek to integrate neural network-based methods with symbolic knowledge-based approaches.

Neuro-symbolic AI is all about bringing together the best of both worlds:

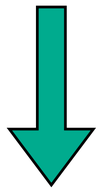
- **Neural networks**: super good at learning from huge amounts of data,
- **Symbolic AI:** top-notch reasoning and handling symbols.

Making AI systems that are stronger, more flexible, and think more like us.



Deduction vs Induction

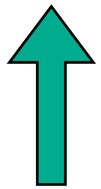
Human reasoning is the synthesis of **deductive** and **inductive** thinking.



Deductive Reasoning: the conclusion is obtained from the premise and inference rules. Starting from a set of general premises and then drawing a specific conclusion that contains no more information than the premises themselves.

Top-down: certain conclusions

Ex: If something is true of a general class, then it is true of the members of the class.



Inductive Reasoning: using specific observations, such as observed patterns, to make a general conclusion. Starts with a set of premises, based mainly on experience or experimental evidence and uses those premises to generalize a conclusion.

Prone to flaws: conclusions drawn using *induction* go beyond the information contained in the premises.

Bottom-up: probabilistic conclusions

Ex: …

Foundations of Neuro-Symbolic AI

The combination of the two main pillars can happen in different ways, and there is currently no single or dominant vision of how this should be done. Various strategies are being explored.

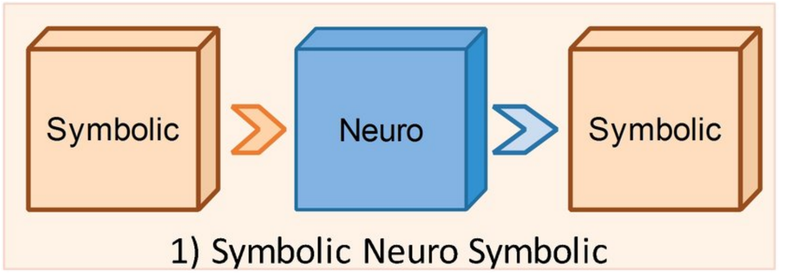
In the absence of a singular approach, many refer to the **taxonomy** developed by **Henry Kautz**, Professor Emeritus at the University of Rochester, in 2020.

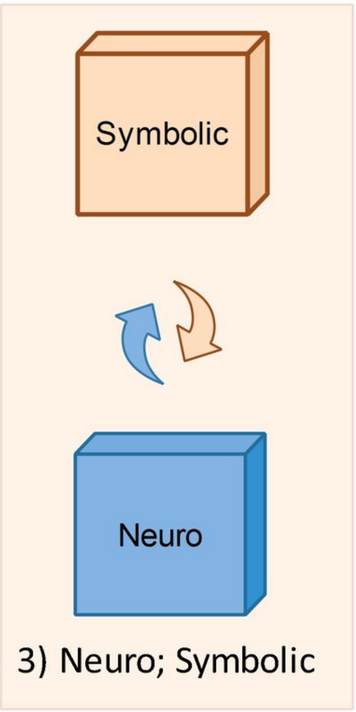
Neuro Symbolic Taxonomies

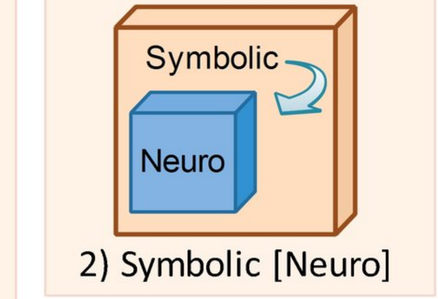
Kautz's Taxonomy

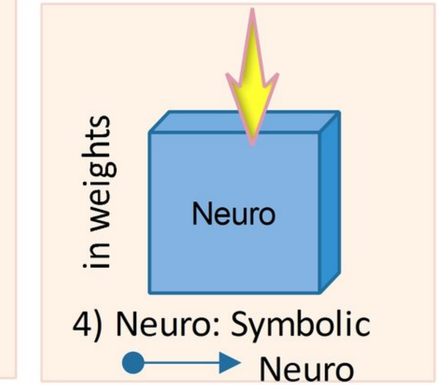
Six types of neuro-symbolic systems:

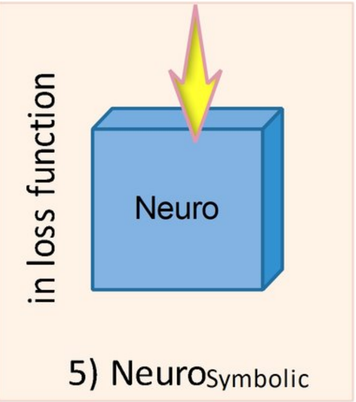
1. Symbolic Neuro symbolic
2. Symbolic [Neuro]
3. Neuro | Symbolic
4. Neuro:Symbolic → Neuro
5. Neuro_{Symbolic}
6. Neuro [Symbolic]

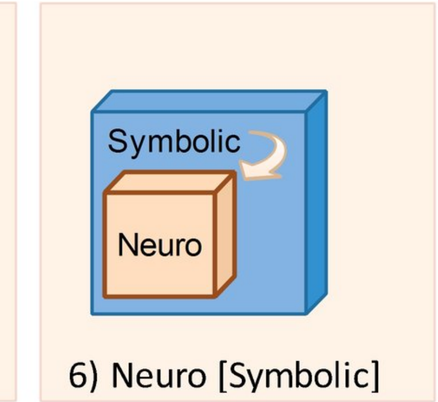




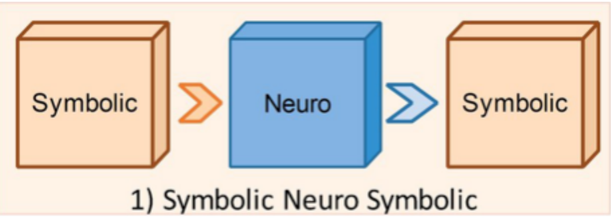






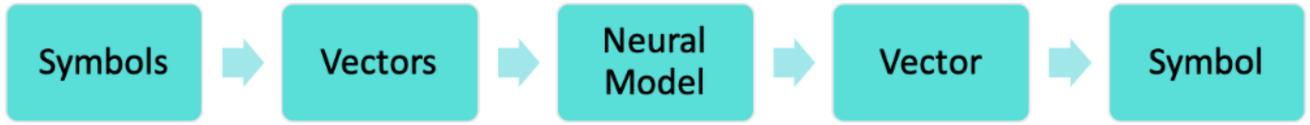


Kautz's Taxonomy



1. Symbolic Neuro Symbolic

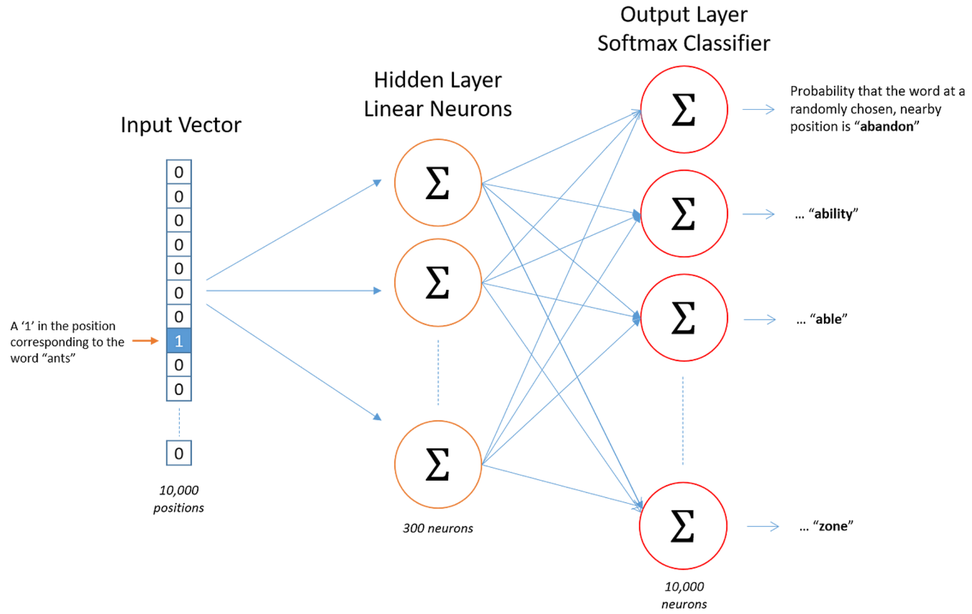
Standard operating procedure of deep learning, where any/all the symbols in the problem are converted to vector embeddings. Vectors are then processed by neural models, which spits out another vector which is then converted to the required symbol



<u>Examples:</u>

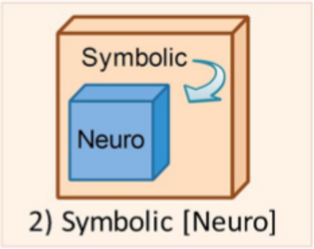
- Natural Language Processing:

words are converted to vectors before the neural manipulations



ChatGPT Gemini Claude

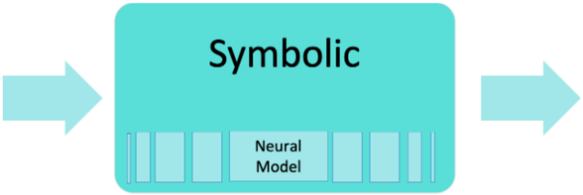
Kautz's Taxonomy



2. Symbolic [Neuro]

Symbolic system with a **neural sub-routine**.

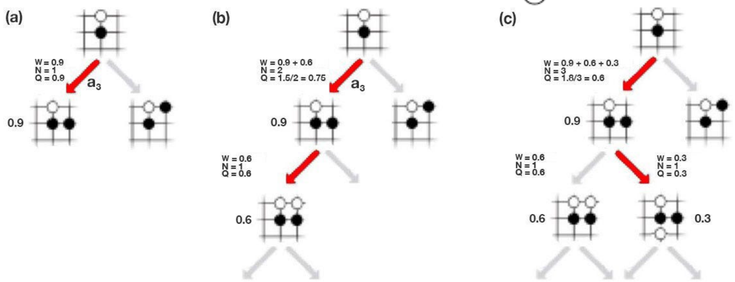
All symbolic solver which uses neural model internally as pattern recognition subroutine for one or more function.

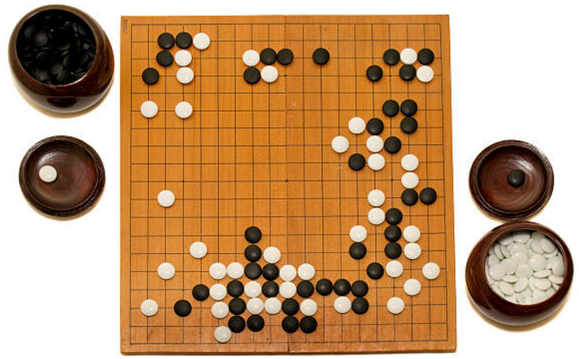


Examples:

- **NeuroCore:** branching heuristics to accelerate SAT solving
- AlphaGo: Monte Carlo search with a neural network used as a heuristic function

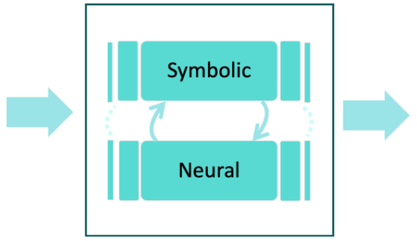
AlphaGo

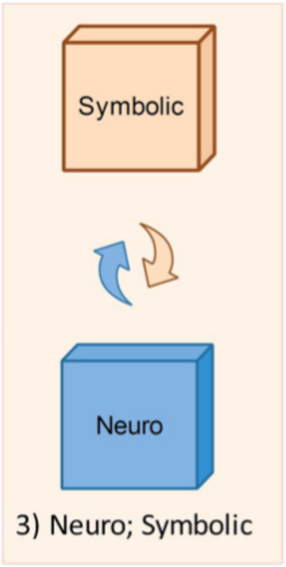




Kautz's Taxonomy

3. Neuro | Symbolic





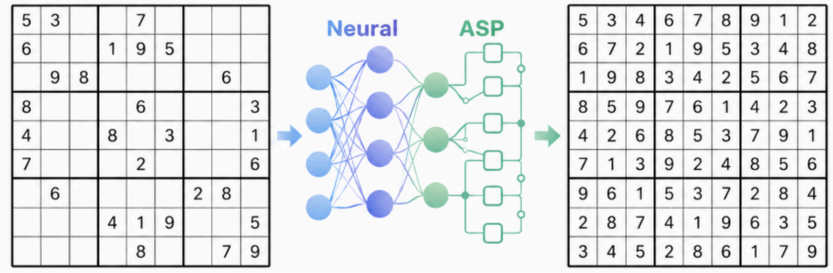
Neural and Symbolic systems are leveraged for different tasks in a big pipeline.

Both systems communicate with each other either to extract information or to improve the individual/collective systems performance.

Neural network converts non-symbolic input (e.g., pixels) into symbolic input that is manipulated by a symbolic reasoning systems.

Differs in that the **neural** component is a **co-routine** instead of a sub-routine.

Examples:



$$\underbrace{\text{get key} \rightarrow \text{open door} \rightarrow \text{pick box} \rightarrow \text{deliver box}}_{\text{Symbolic plan}}$$

$$\pi_\theta(a_t \mid s_t, g)$$

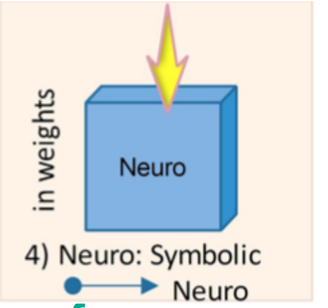
Neural policy conditioned on sub-objective

- **NeurASP:** Neural networks + Answer Set Programs

- Integrating symbolic planners with neural RL agents: both systems feed off of each other.

Kautz's Taxonomy

4. Neuro:Symbolic → Neuro



Symbolic knowledge is compiled **into the structure of neural models** or **into the training process of neural network**.

Deeper coupling relation between neural and symbolic approaches, including **three cases**:

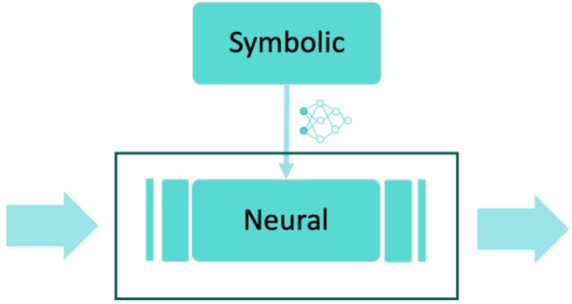
1) symbolic knowledge is compiled into the **training set** (vectorized data) of neural network for training operations;

2) symbolic knowledge is transformed into initial **architecture** and weight set **of neural network**, or logical neural networks are constructed to make the network units correspond to the elements of logical formulas one by one;

3) the symbolic knowledge is mapped to an embedding vector, which acts as the constraint of the neural network **loss function**.

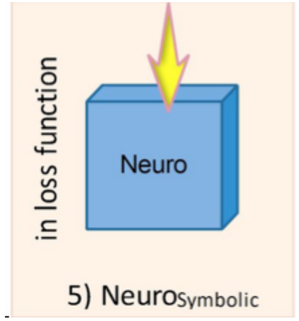
Examples:

- **Logic Neural Networks (LNN):** architecture determined by logical formulas and a loss function that penalizes logical contradictions.
- Differentiable Inductive Logic Programming (δ-ILP)



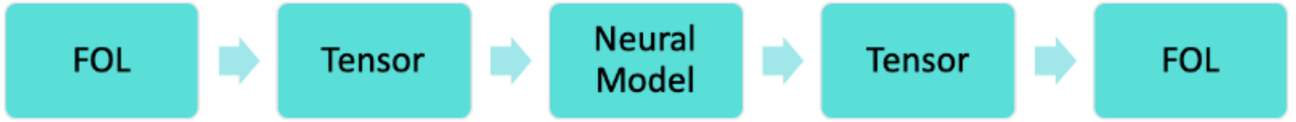
Kautz's Taxonomy

5. Neuro_{Symbolic}



Direct **encoding of logical statements into neural structures** (through the use of embeddings).

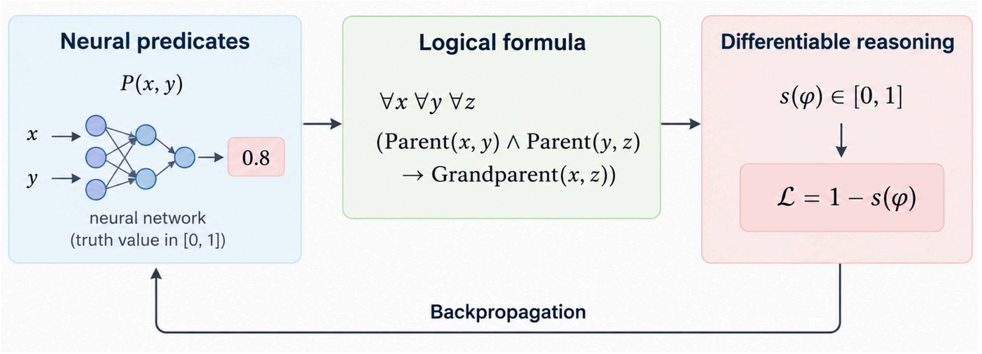
First-order logic language is tensorized and neural methods are used to perform reasoning over this tensorized first-order logic representation. They act as a form of regularization



The difference between this and type 1 is that in type 1 only the objects are represented as vector embeddings. Here complete first-order language is represented by tensor embedding, including object, relation, clauses/rules, functions, predicates.

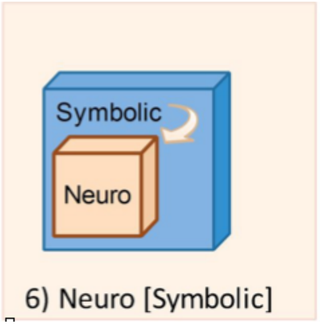
<u>Examples:</u>

- **Logic Tensor Networks (LTN)**
- Deep Ontological Networks



Kautz's Taxonomy

6. Neuro [Symbolic]

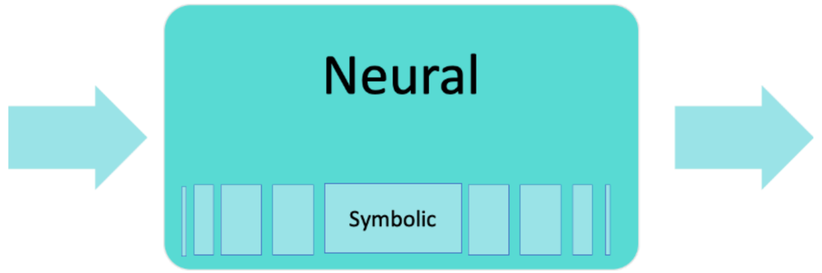


Transpose of Type 2. The overall Neural model performs symbolic reasoning by either learning the relations between the symbols or paying attention to selected symbols at certain point.

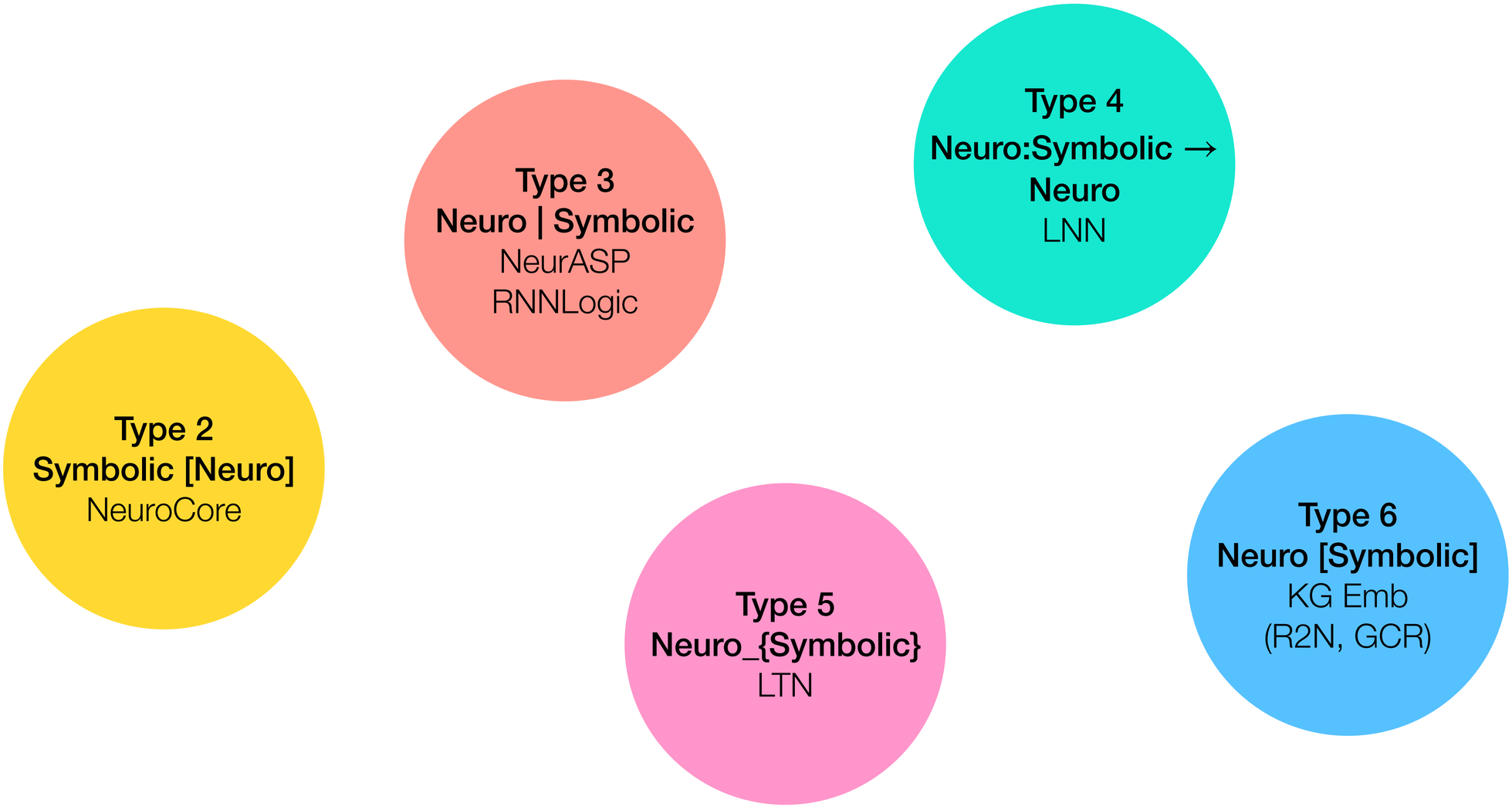
Fully-integrated system that can carry out true **symbolic reasoning within a deep neural network engine.**

Examples:

- **R2N** (Relational Reasoning Networks): Performs neural relational reasoning by propagating information through rule-based structures.
- **GCR** (Graph Collaborative Reasoning): Integrates logical rules into graph-based neural computation for relational reasoning.



Kautz's Taxonomy



NeSy AI & XAI

NeSy AI aims to make reasoning interpretable by design,
while **XAI** aims to make model behavior understandable through explanations.

The main difference is where interpretability comes from.

- **Neuro-Symbolic AI (NeSy):** Interpretability can emerge by design, through explicit symbolic representations, logical rules, and structured reasoning processes.

- **Explainable AI (XAI):** Explainability often comes post hoc, through techniques that explain the predictions or behavior of an already trained model.

Example: Medical diagnosis

NeSy AI: A system reasons using an explicit rule: $\text{Fever}(x) \wedge \text{Cough}(x) \Rightarrow \text{SuspectedInfection}(x)$

The rule can provide a traceable justification for the prediction.

XAI: A neural network predicts an infection, and SHAP identifies fever and cough as influential features.

NeSy AI Interpretability	XAI explainability
Often intrinsic to the model	Often post-hoc
Explains through symbolic knowledge and reasoning	Explains through feature attribution, examples, or approximations
Can expose reasoning steps	Often explains what influenced a prediction
Interpretability depends on how the symbolic and neural components interact	Explanation fidelity depends on the method

Remark: Not all NeSy systems are inherently interpretable (e.g., embedding-based approaches), and not all XAI methods are post-hoc (e.g., inherently interpretable models).

Bibliography

DeLong, L. N., Mir, R. F., & Fleuriot, J. D. (**2024**). **Neurosymbolic AI for reasoning over knowledge graphs: A survey**. *IEEE Transactions on Neural Networks and Learning Systems*

Coming next…

Fundamentals of Logic